HuLa Drone User Guide

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About this guide

This guide covers the HuLa drone hardware itself: its parts, battery, pairing modes, light colors and safe flying. The companion HuLa App Instructor Guide covers the app and its blocks.

The HuLa (by Shenzhen HighGreat Innovation) is a 100 g indoor drone for AI and programming lessons, for ages 14 and up. It has a front camera on a gimbal, a laser emitter and receiver, 4-way infrared obstacle sensing, and a downward camera for optical-flow and QR-code positioning.

Facts come from the official *Hula User Manual* (revision 30 Jan 2024) unless marked as our own classroom observation.

Items the manual refers to (check them against your kit's packing list):

- The drone, with propeller guards fitted
- 1 flight battery (1200 mAh)
- Spare propellers, marked "75 mm" and "75 mmR"
- Propeller removal tool
- Small screwdriver for the obstacle-sensor knobs, and a card pin for the reset hole
- Micro USB cable
- Charging box (optional, sold with some kits)

Parts of the drone

The top shell (lamp cover) carries the status light; the power button, reset hole and sensor knobs sit on the back, under the battery.

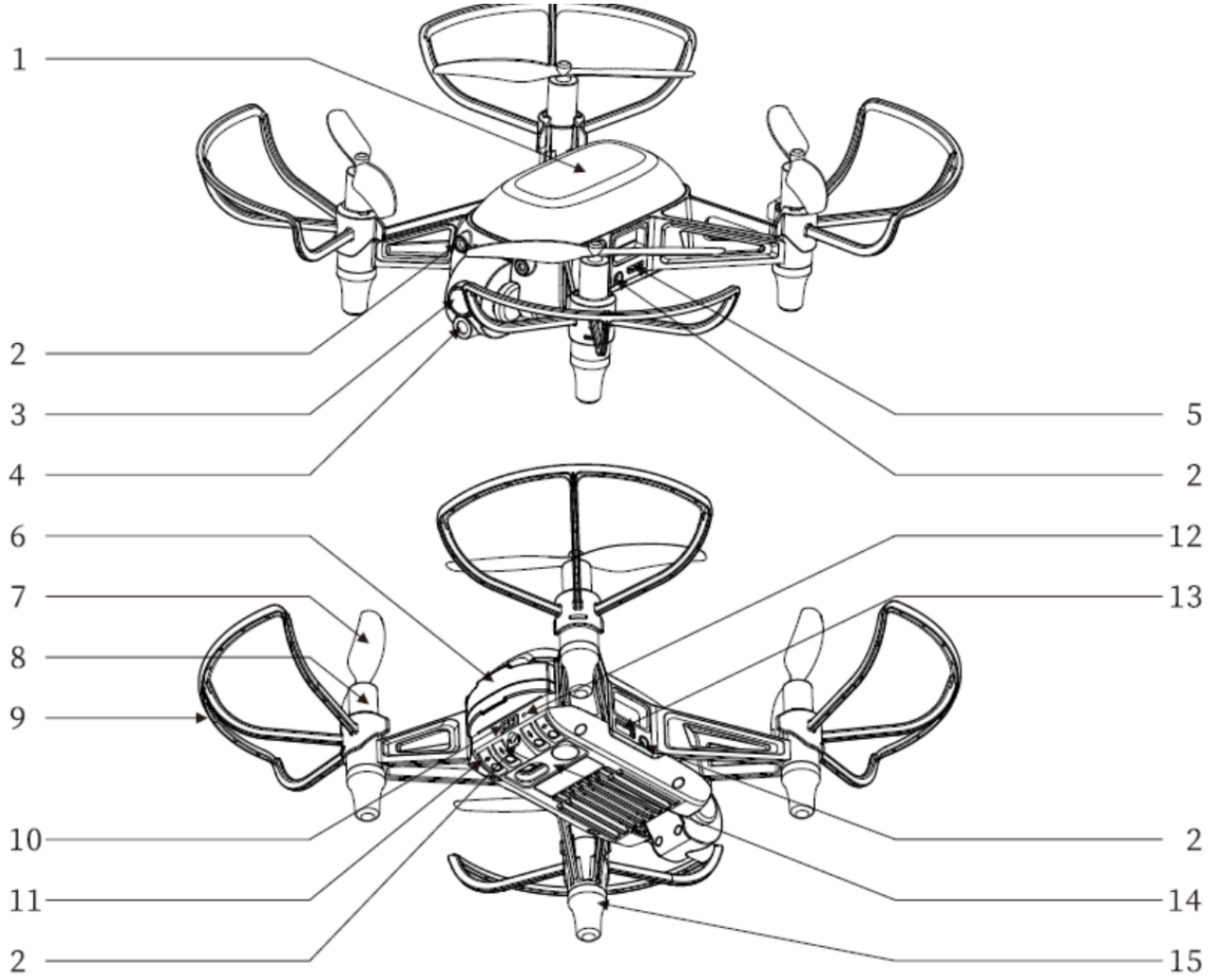


Diagram from the official Hula User Manual, section 1.5.

#	Part	What it does
1	Lamp cover	Status light (see Light colors). Also hides the infrared obstacle receiver: never cover it.
2	Infrared obstacle transmitters	One per side (front, back, left, right). Sense walls about 30–50 cm away.
3	Gimbal / front camera	1080p photos, 720p video, live view, AI recognition. Tilts up and down.
4	Laser transmitter	Red 640 nm laser for battle games. Never aim at eyes.
5	Micro USB port	Data / service port on the drone.
6	Battery	1200 mAh, slides into the back with a click.
7	Propellers	75 mm. Two types: "75 mm" and "75 mmR" (see Propellers below).

#	Part	What it does
8	Motors	Get hot in flight. Don't touch after flying.
9	Propeller guards	Protect props and fingers. Keep them fitted.
10	Power button	Hold 2 s = on/off. Press 3 times fast = switch connection mode.
11	Obstacle adjustment knobs	Four small screws that set the obstacle-sensing distance per direction.
12	Reset button	Inside the reset hole next to the power button. Pressed with a pin.
13	Expansion serial port	For add-on modules.
14	Visual positioning system	Downward camera (optical flow) + ToF height sensor. Reads floor texture and QR mats.
15	Foot mats	Landing feet.

Propellers

The two prop types spin in opposite directions. Fit them in the wrong place and the drone will not take off.

Position (looking from behind, nose forward)	Prop marking
Front-left and rear-right	75 mmR
Front-right and rear-left	75 mm

- To remove a prop, slide the removal tool between the prop cap and the motor and lever it off. Never pull props off by hand.
- When fitting, leave just enough gap under the cap for the removal tool.

Battery and charging

A full battery gives 9–10 minutes of hover (measured with lights off); real lesson flying is shorter, so keep spare charged batteries rotating.

Fitting the battery: hold the drone steady, sticker side of the battery facing down, and push it into the slot on the back until it clicks.

Charging method	Time to full	Light while charging	Light when full
Micro USB cable into the battery, from any 5 V USB charger	about 1 h 40 min	Solid red (on the battery)	Red light off
HuLa charging box (needs a 5 V 3 A adapter)	about 1 h	Solid red (on the box)	Solid green (green also means empty slot)

- Low battery in flight: the drone light slowly flashes red, and by default it lands itself at 10%.
- Advanced moves (flip, bounce, follow, line patrol) need at least 30% battery.
- Below 5 °C the battery runs out noticeably faster.
- The manual suggests replacing a battery after about 30 hours of total flight time (shown in the app's Aircraft status).
- Stop using any battery that is wet, swollen, leaking, smelly or bent. Use only HuLa batteries and never short the contacts.
- Storage: up to 1 month at -20–45 °C, 3 months at -20–30 °C, long-term around 25 °C.

Power on and off

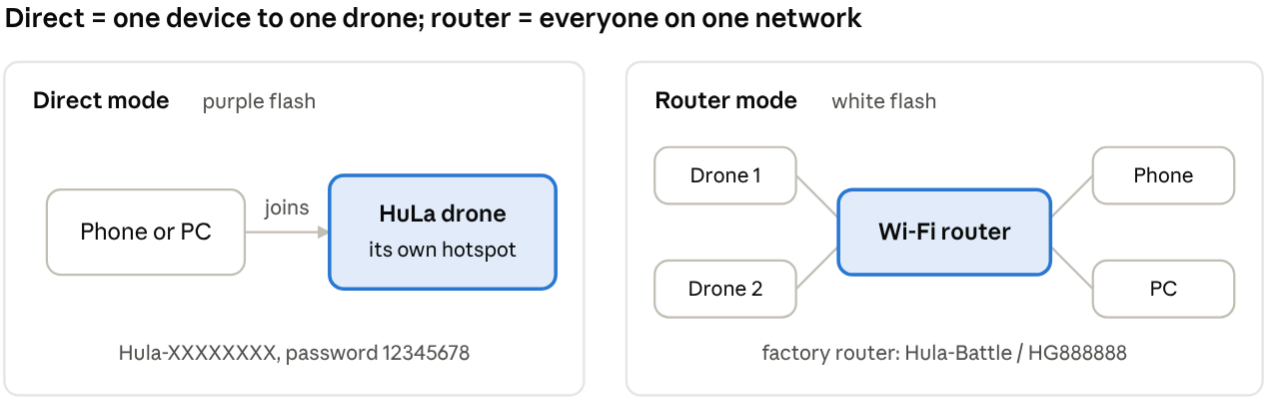
Hold the power button on the back for **2 seconds** to switch on; hold it 2 seconds again to switch off.

1. Place the drone flat on the floor, nose pointing away from you, before switching on.
2. Hold the power button 2 s.
3. The lamp flashes **red, green, blue**: the power-on self-check. Don't move the drone during this.
4. The lamp then flashes **purple** (direct mode) or **white** (router mode). This tells you which connection mode it is in.
5. When the self-check ends it settles into a **breathing** light (slowly fading in and out): green in direct mode, white in router mode. It is now waiting for a connection.

You can also power the drone off from the app (Settings → Power Off), but only once it has landed.

Pairing and connection modes

The HuLa connects over Wi-Fi only: there is no handheld remote. It has two modes, and you switch between them by **pressing the power button 3 times quickly** while it is on. New drones ship in direct mode.



the two connection modes

	Direct connection mode	Router (networking) mode
How it links	Your phone/tablet/PC joins the drone's own Wi-Fi hotspot	The drone and your devices all join the same Wi-Fi router
Light after self-check	Flashing **purple**	Flashing **white**
Waiting (not connected)	Breathing green	Breathing white
Connected	Solid green	Solid white (manual)
Drone's Wi-Fi in the list	Yes: "Hula-XXXXXXXX"	No
Drones per device	One	Many
Use it for	Single driver, single-drone programming, Battle-Melee	Team battles, formations, a class sharing one network, the Python SDK

Switching mode drops any current connection, so switch before you connect.

Method 1: Direct connection (one drone, quickest)

1. Power on (hold 2 s) and wait for the red/green/blue self-check.

2. Check the light: **purple** flashing = direct mode. If it flashes **white**, press the power button 3 times quickly and wait for purple.
3. On your device, open Wi-Fi settings and join **Hula-XXXXXXXX** (the X's are unique to each drone). Default password: **12345678**.
4. Open the HuLa app. When the drone's light turns **solid green** and the app says connected, you are paired.

Can't see the Hula Wi-Fi? Refresh the Wi-Fi list, switch mode again (3 presses), or restart the drone.

Method 2: Router mode (many drones, one network)

Each drone stores one router name and password. Out of the factory it looks for a router called **Hula-Battle** with password **HG888888**. You can either set your router to that, or teach each drone your own router.

Option A – rename the router (easiest for a class kit):

1. Set your router's Wi-Fi name to **Hula-Battle** and password to **HG888888** (see your router's manual). Note: other devices on that router will need reconnecting.
2. Switch each drone to router mode (3 presses, white flashing). Each one joins the router by itself.
3. Connect your phones/PCs to the same Hula-Battle Wi-Fi and open the app.
4. First time only: in the app, open **Device** (equipment base) and pick your drone from the list. Each drone is listed by its unique SN code.

Option B – give the drone your router's details:

1. Put the drone in direct mode and connect to it (Method 1).
2. In the app's connection guide, send your router's Wi-Fi name and password to the drone.
3. Switch the drone to router mode (3 presses). It finds and joins your router.
4. Repeat steps 1–3 for every drone.

If the drone can't reach the router after switching, the light flashes **red for 3 seconds**, then goes back to breathing white.

Method 3: Python SDK (PC programming)

The Python SDK in this project only works in router mode; it does not work over the drone's own hotspot.

1. Activate the drone once with the HuLa phone app (this can't be done from a PC).

2. Use the phone app to give the drone your Wi-Fi details (Option B above) and switch it to router mode.
3. Find the drone's IP address in your router's list of connected devices (it shows as something like **hg-<id>**). Pass that address to the SDK.
4. Tip: give the drone a fixed (reserved) IP in the router, as the address can change after a router restart.

Classroom note: on our test unit the light showed solid **green**, not white, once joined to a router. Treat either as "connected" and confirm in the app.

Changing or resetting the drone's Wi-Fi

- **Rename the hotspot or change its password:** connect directly, then in the app go to Settings → System → Modify the aircraft WiFi. The drone restarts afterwards.
- **Switch hotspot band (2.4 GHz / 5.8 GHz):** Settings → System → WiFi frequency band, while directly connected. Re-enter the password on the new network.
- **Forgot the password? Factory-reset the Wi-Fi:** push a pin into the reset hole (next to the power button) and hold **5 seconds**. A fast-blinking green light confirms. The hotspot goes back to Hula-XXXXXXXX / 12345678.
- **Roll back the firmware:** hold the reset button **10 seconds**. The drone restarts on its previous firmware version. Don't press the reset button hard.

Light colors and patterns

The lamp on top of the drone tells you its state at a glance. Read the **color** and the **pattern** together: *solid* = steady on, *breathing* = slowly fading in and out, *flash* = blinking.

Color	Pattern	Meaning	What to do
Red + green + blue	Flash	Power-on self-check	Wait; keep the drone still and flat
Purple	Flash	Direct mode (after self-check, or after switching to it)	Join the Hula-XXXXXXXX Wi-Fi
White	Flash	Router mode (after self-check, or after switching to it)	Make sure the router is on
Green	Breathing	Direct mode, positioning OK, nothing connected yet	Connect the app

Color	Pattern	Meaning	What to do
White	Breathing	Router mode, positioning OK, nothing connected yet	Connect the app on the same router
Green	Solid	Connected (direct mode)	Ready to fly
White	Solid	Connected (router mode)	Ready to fly
Green	Fast flash	Wi-Fi reset to factory settings	Reconnect with the default password
Green	Breathing (in formation)	Controlled by the app over a router, formation mode	—
Green	Solid (in formation)	Take-off check passed: steps, firmware, number, battery and hardware all OK	Ready for the show
Blue	Flash	Data transfer: firmware upload, dance steps, or pairing finishing	Don't switch off; wait for it to return to normal
Blue	Double flash, repeating	Infrared sensor sees an obstacle	Move the drone away from the wall/object
Blue	Breathing	IMU calibration in progress (or formation take-off check)	Follow the app's calibration steps
Orange	Breathing	Special flight mode running (fly-around, bounce, line patrol, follow)	Returns to normal when the move ends
Orange	Flash	Searching for an object to recognise (or selected by the host in formation)	Keep the target in camera view
Orange	Solid, 5 s	Object recognised	—
Orange	Flash, 3 s	One IMU calibration step done	Continue to the next step
Red	Slow flash	Low battery	Land and change the battery

Color	Pattern	Meaning	What to do
Red	Solid	Hardware fault or failed self-check (also: failed calibration, or drone tilted during calibration)	Restart; if it stays red, recalibrate the IMU or contact support
Red	Flash, 3 s	Couldn't join the router after switching to router mode	Check the router name and password
Red	Breathing	Formation take-off check failed	Check firmware, numbering and battery
(any)	Breathing	Positioning problem: the floor camera can't see well	Better light, textured floor; clean the bottom camera
Off	—	Lights turned off in settings or by a program. Warning lights still show	—

During a firmware upgrade the lights go off, then flash red/green/blue as the drone reboots. That is normal.

Your programs can also set the lamp to any color (the light blocks in the app, or `set_light` in the SDK), so a custom color during a program run is not a status signal.

Before you fly

The HuLa holds its position by watching the floor, so the room matters as much as the drone.

Flight area

- Indoors, good even lighting, no strong sunlight or spotlights on the floor.
- A floor with clear texture or pattern (a HuLa mat, tiles, carpet with a pattern). Plain single-color floors, mirrors, glossy or wet floors make it drift.
- No fans or air-con blowing on it (it can't fly in wind of 3 m/s or more).
- Everyone at least **2 m** away from a flying drone.
- Open space: keep more than 50 cm from walls so obstacle sensing doesn't push it around.

Checklist

- [ ] Battery charged and clicked in

- [ ] Propellers: correct type in each position (75 mmR front-left/rear-right), not chipped, guards fitted
- [ ] Bottom camera and ToF sensor clean and uncovered
- [ ] Lamp cover not covered (the obstacle receiver is inside it)
- [ ] Drone placed flat, nose away from you
- [ ] Light shows connected (solid green / white) and the app agrees
- [ ] Aircraft status in the app is green (ToF and optical flow showing red **on the ground** is normal)

Calibrating

- **IMU calibration** (accelerometer and gyroscope): in the app, Settings → Control → IMU Calibration. Do it if the drone drifts or tilts on its own, or after a crash. Follow the on-screen positions; the light breathes blue, flashes orange after each step, and flashes green when done. Red = hold it level and try again.
- **Obstacle sensing distance:** factory set to about 30–50 cm. To change it, turn the four small knobs on the back with the supplied screwdriver: clockwise = shorter distance, counter-clockwise = longer. Adjust facing a white board in normal room light. Turn gently.
- **Laser crosshair** (for battles): in Battle → Melee → Settings, fire the laser at a white wall and tap the red dot so the crosshair lines up.
- **Camera level:** Settings → Camera Reset puts the gimbal back to horizontal.

Flying

Take-off goes to **1 m** by default; the drone flies best between 1 and 4 m and can go from 0.5 m up to 10 m.

Action	How
Take off	In the app, press and hold the take-off button for 1 s
Land	Press land, then confirm
Pause a move	Press and hold the joystick (any direction) for 1 s
Lost connection	The drone waits 10 s, then lands where it is
Low battery	Lands by itself at 10% (can be turned off in Settings → Control, not recommended)

Action	How
Stop a program	Press Stop in the programming screen: the drone lands and the program resets

Speeds: the app's Movement Speed setting is Low 0.5, Medium 1 or High 2 m/s; 0.5–1 m/s is best for programs. Below 0.5 m height, keep it under 0.2 m/s or it may crash.

Safety rules

- Never hold the drone in your hand while it flies.
- Stay 2 m away during take-off and landing.
- Don't close the app, leave the flight screen or drop the Wi-Fi mid-flight.
- Motors and the heatsink underneath get hot. Let them cool before handling.
- Video recording stops after 8 minutes per clip.

Sensors and built-in features

Feature	What it is	Tips
Front camera on gimbal	1920×1080 photos (JPG), 720p 30 fps video (MP4), 71° view; live view in the app	Tilt it level for QR following. Set anti-flicker to 50 or 60 Hz to match your room lights.
Optical-flow camera + ToF (underneath)	Holds position and height by watching the floor. Accurate to about ±20 cm sideways, ±10 cm height	Needs light and floor texture. Keep the lens clean.
QR-code positioning	Reads HuLa QR mats on the floor for precise position: about ±5 cm sideways, ±10 cm height	Mats must be flat, unfolded and well lit.
Infrared obstacle sensing	4 directions, about 30–50 cm range; blue double-flash when it sees something	Turn on with Settings → Control → Obstacle Avoidance. Dark or shiny objects are detected at shorter range.
Laser emitter and receiver	Red 640 nm laser for battle games; the receiver registers hits from other drones	Never aim at eyes. Receiving can be turned off (Settings → Control → Infrared Laser Reception).

Feature	What it is	Tips
Lamp	RGB status light, up to 1.5 W, brightness adjustable; programmable color	Lights on shorten flight time slightly.
Line patrol	Follows a line on the floor	Line wider than 5 cm, strong contrast (light line on dark floor best), turns no sharper than 45°, fly at 1–2.2 m. Stops if no line is found for 30 s. Video drops to 360p.
QR / color follow	Follows a QR code or a color held in front of the camera	Fly at 1–2.2 m, code 1–2.2 m away, move it slowly (under 0.3 m/s sideways).
One-click moves	Flip, bounce, vertical circle, fly-around (circles a point 1.5 m ahead), spin, curve	Need 30%+ battery, 0.5 m+ height and open space.
Storage	Photos and videos are saved on the drone; the app's Album downloads them	Check free space in Aircraft status.

Troubleshooting

Problem	Likely cause	Fix
Hula-XXXXXXXX Wi-Fi doesn't appear	Drone is in router mode (white flash)	Press power 3 times for purple, refresh the Wi-Fi list, or restart the drone
Wrong Wi-Fi password	Password was changed	Hold reset 5 s with a pin; default is 12345678
App won't connect in router mode	Device on a different network, or drone doesn't have the router details	Put everything on the same router; resend the router details (Pairing, Option B)
Red flashes 3 s after switching to router mode	Router not found or wrong password	Check the router name/password; for factory settings the router must be Hula-Battle / HG888888

Problem	Likely cause	Fix
Drone won't take off	Props in the wrong positions, low battery, or a red status item	Check prop markings; charge; read Aircraft status in the app
Drifts or wobbles in the air	Plain/shiny floor, poor light, dirty bottom camera, or IMU needs calibrating	Use a textured mat and good light; clean the lens; run IMU calibration
Bounces away from walls	Obstacle sensing reacting	Fly more centrally, or shorten the distance with the knobs
Solid red light	Hardware fault or self-check failed	Restart on a flat surface; recalibrate IMU; if it persists contact support
Slow red flash	Low battery	Land and swap the battery
Flight time much shorter than usual	Cold battery, lights on, or old battery (30 h+)	Warm the battery to room temperature; replace old batteries
Flickering live video	Room lights' frequency	Change anti-flicker to 50 or 60 Hz
Python SDK can't connect	Drone in direct mode, or wrong IP	Use router mode; read the drone's IP from the router (hg-<id>)
Firmware upgrade failed	Connection dropped during upload	Keep the app open and the drone on; press Re-upgrade. If the drone misbehaves after, hold reset 10 s to roll back

For repairs, contact your dealer or HighGreat service: service@hg-fly.com.

Specifications and quick reference

Quick reference

Do this	How
Power on / off	Hold power button 2 s
Switch direct ↔ router mode	Press power button 3 times quickly

Do this	How
Reset Wi-Fi to factory	Pin in reset hole, hold 5 s (green fast flash)
Roll back firmware	Pin in reset hole, hold 10 s (drone restarts)
Drone hotspot	Hula-XXXXXXXX / 12345678
Factory router details	Hula-Battle / HG888888

Specifications

Item	Value
Weight (with battery)	100 g ± 3 g
Size	189.3 × 184.6 × 50 mm, 128 mm motor-to-motor
Propellers	75 mm (3 in)
Max speed	3 m/s sideways, 1.2 m/s up, 1 m/s down
Max height	10 m
Max tilt	20°
Range	50 m
Wi-Fi	2.4 GHz (2.412–2.462) and 5.8 GHz (5.745–5.825)
Flight time	9–10 min (hover, lights off)
Operating temperature	0–40 °C
Battery	1200 mAh, 3.8 V lithium-ion, 31 g
Camera	1920×1080 photo, 720p 30 fps video, 71° field of view
Laser	640 nm
Charging box input	5 V 3 A, 1.4 A charging current
Apps	iOS 15+, Android 11+, Windows 10+ (64-bit)
Recommended age	14+, younger with adult supervision

Sources

- *Hula User Manual*, HighGreat Innovation, revision 30 January 2024 (Training/Hula User manual.pdf)
- HulaSDK README, network setup notes (hands-on testing with this project's drone)